

NetCfg: A Declarative Network Configuration Compiler

Technical Reference

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Abstract. NETCFG is a Rust-based network configuration compiler that transforms high-level topology intent into verified, target-specific device configurations. The compiler processes intent through a five-layer pipeline: (1) Intent Transformation via composable primitives, (2) Topological Validation via a DSL-driven assertion engine, (3) Lowering to a vendor-neutral Device Intermediate Representation (DeviceIR), (4) Target-Specific AST Transformation (TSDM) for hardware-aware lowering, and (5) Syntax Serialization via dumb templates. This report documents the data model, the transformation DSLs, and the AST-based lowering mechanism that decouples architectural logic from physical hardware constraints.

Version	Date	Changes
1.0	March 2026	Initial release (v1.3 feature set)
1.1	March 2026	Updated to v2.0: policy primitives, WASM plugins, LSP, REST API, 4 new CLI commands

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1 Introduction

What NETCFG is

NETCFG is a command-line tool and Rust library that compiles declarative network topology descriptions into per-device configuration files. It takes a *blueprint* (describing what the network should look like) and a *mapping* (describing how to extract vendor-specific configuration from the topology) and produces deterministic, diffable, auditable configuration artifacts for every device in the network.

NETCFG replaces the manual “glue” layer between source-of-truth systems (NetBox, Nautobot), automation frameworks (Ansible, Nornir), and network modelling tools (Batfish). It treats network configuration as a *compilation problem*: topology intent is compiled through a typed pipeline, not interpolated through ad-hoc templates.

Design philosophy

Two principles explain every design decision in this document:

1. **Lowering over templating.** The operator’s intent is lowered through successive stages of abstraction. It moves from a high-level graph to a neutral AST (DeviceIR), then to a platform-specific AST (TSDM), and only finally to CLI syntax. This ensures that hardware quirks and vendor-specific logic are handled via structured AST transformations, not ad-hoc string concatenation.
2. **Separation of what from how.** The blueprint declares *what* the network should look like (topology transformations). The mapping declares *how* to extract device configuration from the result. The same blueprint can produce Cisco IOS, Arista EOS, or Juniper JunOS output by swapping only the mapping and templates.

Audience

This document is intended for:

- Engineers integrating NETCFG into a network automation pipeline.
- Contributors to the NETCFG and NTE codebases.
- Researchers evaluating graph-based approaches to network configuration.

Familiarity with YAML, basic graph theory, and IP networking is assumed. Experience with Rust is helpful but not required for the API sections.

How to read this document

Readers wanting to get started quickly should read §2 (data model), §3 (blueprint DSL), and §6 (CLI reference), then refer to individual sections as needed. §4–§5 cover the compilation pipeline and core mechanisms in depth and can be deferred until required. §7 provides a complete end-to-end worked example that traces a blueprint through every pipeline stage. §9 documents the language server and REST API for editor and CI/CD integration.

2 Background: Network Configuration as Code

NETCFG models the network as a directed graph $G = (V, E, \tau, \lambda, D)$ where V is a set of nodes (devices, protocol instances), $E \subseteq V \times V$ is a set of edges (links, peering sessions), $\tau : V \rightarrow T$ assigns each node a *type* (Router, Switch), $\lambda : V \rightarrow L$ assigns each node a *layer*, and $D : T \rightarrow DataFrame$ maps each type to a columnar property store (Polars DataFrame).

2.1 Layers

A layer represents a plane of the network: `physical` for the hardware topology, `bgp` for BGP peering sessions, `ospf` for OSPF adjacencies. A single physical router (node 1, layer `physical`) may have corresponding protocol nodes (node 101, layer `bgp`; node 201, layer `ospf`), each linked to node 1 via a *parent mapping* (`_source_id` in the node’s `data_json`).

Layers allow the compiler to model the physical and logical topologies simultaneously, with cross-layer references preserving traceability from a protocol node back to the physical device that hosts it.

2.2 Per-node state: DeviceContext

Each node's properties are stored as a JSON blob (`data_json`) in a Polars DataFrame keyed by node type. NETCFG deserialises this blob into a typed `DeviceContext` struct:

Listing 1: DeviceContext: the per-node state model.

```
pub struct DeviceContext {
    pub hostname: String,
    pub device_os: Option<String>,           // template selector
    pub management_ip: Option<String>,
    pub src_ip: Option<String>,              // written by provision_ips
    pub dst_ip: Option<String>,              // written by provision_ips
    pub subnet: Option<String>,              // written by provision_ips
    pub src_ipv6: Option<String>,            // dual-stack support
    pub dst_ipv6: Option<String>,
    pub subnet_v6: Option<String>,
    pub vrf: Option<String>,                 // VRF domain
    pub peer_interfaces: Option<HashMap<String, String>>,
    pub next_interface_index: Option<u32>,
    pub stanzas: Vec<Stanza>,                // accumulated config units
    pub metadata: HashMap<String, JsonValue>, // extension point
}
```

Primitives progressively enrich this context: `mesh_nodes` establishes edges and writes `mesh_type/peer_count` into metadata; `provision_ips` writes IP fields; `build_protocol_layer` deep-merges config overlays into the cloned node's metadata. By the time rendering occurs, each node's `DeviceContext` contains all information needed to produce its configuration file.

Note

The metadata field uses `serde(flatten)`, so any JSON key not matching a named field is captured here. This makes `DeviceContext` extensible without schema changes.

2.3 Graph engine dependency

The underlying graph is implemented by the NTE topology engine, which provides a petgraph-backed directed graph with Polars DataFrames for columnar property storage. NETCFG depends on `n-te_topology` for the graph and `ank_n-te` for the mapping evaluator and `DeviceIR` types.

3 Blueprint DSL: Declarative Topology Transformations

A blueprint is a YAML document declaring what topology transformations to apply. It is the “what” of the network design.

3.1 Schema and layer ordering

Listing 2: Blueprint top-level schema.

```
pub struct Blueprint {
    pub version: u32,           // must be >= 1
    pub imports: Vec<String>,   // relative paths to merge
    pub layers: Vec<LayerSpec>, // ordered list of layer blocks
    pub assertions: Vec<AssertionSpec>, // post-execution checks
}

pub struct LayerSpec {
    pub name: String,           // non-empty
    pub requires: Vec<String>,  // must name earlier layers
    pub primitives: Vec<Primitive>,
}
```

Three validation passes run at parse time:

1. **Schema validation** via `serde_valid`: field types, non-empty names, integer ranges. Unknown YAML keys are rejected via `deny_unknown_fields`, catching typos at parse time.
2. **CEL selector compilation**: each selector string is compiled into an evaluable CEL program, surfacing syntax errors before execution.
3. **Layer ordering validation**: a forward scan ensures every requires entry names a previously-declared layer. This catches ordering bugs at parse time.

The optional `imports` field lists relative file paths to other blueprint files. At parse time, layers and assertions from imported files are merged into the importing blueprint, enabling reusable assertion libraries (e.g. a standard linting ruleset shared across projects).

Warning

Duplicate layer names are permitted. Multiple `LayerSpec` blocks with `name: input` are common in practice—each block adds primitives to the same layer. The `requires` mechanism enforces inter-layer ordering, not intra-layer uniqueness.

3.2 Selector DSL (CEL surface syntax)

Selectors identify which nodes or edges a primitive operates on. The surface syntax is a lightweight wrapper around Google's Common Expression Language (CEL):

Listing 3: Selector syntax examples.

```
# Select all nodes
selector: "all()"

# Select nodes by property
selector: "nodes[site='a']"
selector: "nodes[role=='spine' && layer=='input']"
selector: "nodes[type=='Router' and site='dc1']"

# Select edges
selector: "edges[src>=0]"
selector: "edges[true]"
```

The selector parser:

1. Extracts the target (nodes or edges) and the filter expression from the bracket syntax.
2. Normalises operators: single `=` becomes `==`; and/or/not become `&&/||/!`; single `&/|` become doubled.
3. Compiles the normalised expression as a CEL program.

At evaluation time, the selector iterates over all nodes (or edges) in the topology, binds each entity's properties as CEL variables (`id`, `type`, `layer`, plus all `data_json` fields), and evaluates the CEL program. Entities for which the program returns `true` are included in the match set.

Note

Undeclared variable references (e.g., querying a field that some nodes lack) silently evaluate to `false` rather than producing an error. This allows heterogeneous topologies where not all nodes have the same properties.

For IP-valued fields, the selector also binds a `{field}_len` variable containing the prefix length, enabling selectors like `nodes[subnet_len == 30]`.

3.3 Primitive catalogue

NETCFG provides twelve primitive types. All primitives follow the *Selector* $\rightarrow$ *Execute* $\rightarrow$ *Mutate* pattern and the *two-pass borrow* discipline: collect all reads into local vectors (immutable borrows), then apply all mutations (mutable borrows), satisfying Rust’s ownership rules without unsafe code.

Table 1 summarises the catalogue. Full field tables follow.

Table 1: Primitive catalogue overview.

Primitive	Purpose
mesh_nodes	Create edges between selected nodes (full, hub-and-spoke, route-reflector)
provision_ips	Allocate IP subnets to edges from a CIDR pool
build_protocol_layer	Clone nodes into a named protocol overlay layer
allocate_resources	Generic pool allocation for ASNs, VLANs, etc.
global_pool	Register a named IP pool for hierarchical carving
global_resource_pool	Register a named resource pool (non-IP)
conditional	CEL-based branching within blueprints
custom_primitive	Named reusable primitive group with parameters
inject_secrets	Read environment variables into node metadata
build_routing_policy	Attach routing policy stanzas (route-maps) to matched nodes
build_access_policy	Attach access-control policy stanzas (ACLs) to matched nodes
wasm_plugin	Execute a WebAssembly plugin binary against matched nodes

3.3.1 mesh_nodes

Creates edges between selected nodes. Supports three mesh topologies: *full* ($\binom{n}{2}$ edges), *hub_and_spoke* (hubs full-meshed, spokes connected to all hubs), and *route_reflector* (same topology as hub-and-spoke). Idempotent: existing edges are detected via a `HashSet<(i32, i32)>` and skipped.

Table 2: mesh_nodes fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for nodes to mesh
mesh_type	String	Yes	full, hub_and_spoke, or route_reflector
hub_selector	String	H&S	Selector for hub nodes
spoke_selector	String	H&S	Selector for spoke nodes
edge_properties	JSON	No	Properties for created edges (deferred; see §10.1)
naming_strategy	String	No	Interface naming: eth, cisco_ge, junos_ge

Example: full mesh within a site

```
- type: mesh_nodes
  selector: "nodes[site='a']"
  mesh_type: full
  naming_strategy: eth
```

With 4 nodes at site A, this creates $\binom{4}{2} = 6$ edges and assigns interface names `eth0`, `eth1`, etc. to each node’s `peer_interfaces` map.

3.3.2 provision_ips

Allocates subnets from a CIDR pool to edges. See §5.1 for the full five-pass algorithm.

Table 3: provision_ips fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for edges
pool	String	Yes	CIDR (e.g. 10.0.0.0/24) or global pool name
subnet_size	u8	No	Prefix length per edge (default: 30 for IPv4, 126 for IPv6)
ipv6_pool	String	No	IPv6 CIDR or global pool name for dual-stack
ipv6_subnet_size	u8	No	IPv6 prefix length per edge
strategy	String	No	dense (default), sparse, or hash
pool_size	u32	No	When referencing a global pool, size of block to carve
vrf	String	No	VRF domain (default: default)

3.3.3 build_protocol_layer

Clones physical nodes into a named protocol layer. See §5.2 for the overlay construction mechanism.

Table 4: build_protocol_layer fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for source-layer nodes
layer	String	Yes	Target layer name (must differ from source)
config	JSON	Yes	Config overlay deep-merged into cloned nodes
clone_underlying	bool	No	Also clone edges between selected nodes (default: false)

3.3.4 allocate_resources

Generic pool allocator for non-IP resources (ASNs, VLANs, loopback IDs).

Table 5: allocate_resources fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for nodes
resource_type	String	Yes	Key name in DeviceContext.metadata
pool	String	Yes	Range (e.g. 65000-65999) or global pool name
strategy	String	No	dense (default)
pool_size	u32	No	When referencing a global pool, number of values to carve

3.3.5 global_pool

Registers a named IP address pool at blueprint scope, enabling hierarchical pool carving across multiple provision_ips primitives.

Table 6: global_pool fields.

Field	Type	Required	Description
name	String	Yes	Pool identifier referenced by provision_ips
pool	String	Yes	CIDR range (e.g. 10.0.0.0/8)
vrf	String	No	VRF domain (default: default)

3.3.6 global_resource_pool

Registers a named non-IP resource pool at blueprint scope.

Table 7: global_resource_pool fields.

Field	Type	Required	Description
name	String	Yes	Pool identifier
resource_type	String	Yes	Resource category (e.g. asn)
pool	String	Yes	Value range (e.g. 65000-65999)

3.3.7 conditional

CEL-based branching. The condition is evaluated as a CEL expression; if true, then_primitives execute, otherwise else_primitives (if present) execute.

Table 8: conditional fields.

Field	Type	Required	Description
condition	String	Yes	CEL expression
then_primitives	Vec<Primitive>	Yes	Primitives if condition is true
else_primitives	Vec<Primitive>	No	Primitives if condition is false

3.3.8 custom_primitive

A named, reusable group of primitives with parameters. The parameters map is pushed onto a stack and accessible within the nested primitives.

Table 9: custom_primitive fields.

Field	Type	Required	Description
name	String	Yes	Primitive name (for reporting)
parameters	HashMap<String, JsonValue>	Yes	Named parameters
primitives	Vec<Primitive>	Yes	Nested primitive list

3.3.9 inject_secrets

Reads environment variables into node metadata. Keys in the secrets map are metadata field names; values are environment variable names.

Table 10: inject_secrets fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
secrets	HashMap<String, String>	Yes	{metadata_key: env_var_name}

3.3.10 build_routing_policy

Attaches routing policy stanzas to matched nodes. Each statement defines a route-map entry with match conditions and set actions, stored as a Stanza in DeviceContext.stanzas for downstream template rendering.

Table 11: build_routing_policy fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
name	String	Yes	Policy name (e.g. BGP-EXPORT)
statements	Vec<RoutingPolicyStatement>	Yes	Ordered route-map entries

Each RoutingPolicyStatement has fields: name (sequence number), action (permit or deny), optional match_condition, and optional set_action. Templates access stanzas via `device.stanzas | selectattr('kind', 'equalto', 'routing_policy')`.

Example: BGP export policy

```
- type: build_routing_policy
  selector: "nodes[role='border']"
  name: "BGP-EXPORT"
  statements:
    - name: "10"
      action: permit
      match_condition: "prefix-list CUSTOMER"
      set_action: "local-preference 100"
    - name: "20"
      action: deny
```

3.3.11 build_access_policy

Attaches access-control policy stanzas to matched nodes. Each rule defines a firewall or ACL entry with source, destination, protocol, and port fields.

Table 12: build_access_policy fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
name	String	Yes	Policy name (e.g. EDGE-ACL)
rules	Vec<AccessPolicyRule>	Yes	Ordered ACL entries

Each AccessPolicyRule has fields: name, action (permit/deny), optional source, destination, protocol, and port. Both policy primitives enforce `SelectorTarget::Nodes`—policies cannot target edges.

Example: DMZ access control list

```
- type: build_access_policy
  selector: "nodes[zone='dmz']"
  name: "UNTRUSTED-TO-INTERNAL"
  rules:
    - name: "100"
      action: deny
      protocol: tcp
      destination: "10.0.0.0/8"
    - name: "999"
      action: permit
```

3.3.12 wasm_plugin

Executes a WebAssembly plugin binary against matched nodes. Requires the wasm Cargo feature flag (`-features wasm`); without it, the pipeline returns a descriptive error. Uses the wasmtime v18 runtime.

Table 13: wasm_plugin fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
plugin_path	String	Yes	Filesystem path to compiled .wasm binary

The execution pipeline: (1) load and compile the .wasm module, (2) instantiate per matched node without host imports, (3) resolve the `apply_node` exported symbol, (4) pass the node's `DeviceContext` as serialised JSON. The current implementation validates compilation and symbol resolution; full linear memory I/O for JSON exchange is planned for a future release (see §11).

Note

The wasm feature is optional to avoid the wasmtime dependency (which adds ~40 MB to the binary) for users who do not need plugin extensibility.

3.4 Design-rule assertions

Blueprints may declare assertions: post-execution invariants checked after all primitives complete but before commit. Each assertion has a severity: `error` (pipeline failure) or `warning` (diagnostic only).

Listing 4: Design-rule assertions in a blueprint.

```

assertions:
- name: "all routers have hostname"
  severity: error
  select: all()
  check:
    field_exists: hostname
- name: "IPs within allocation"
  severity: error
  select: "nodes[src_ip != null]"
  check:
    field_in_cidr:
      field: src_ip
      cidr: "10.0.0.0/8"
- name: "unique hostnames per site"
  severity: warning
  select: all()
  check:
    unique_per_group:
      field: hostname
      group_by: site
- name: "private ASN range"
  severity: warning
  select: "nodes[asn != null]"
  check:
    custom_cel:
      expression: "asn >= 64512 && asn <= 65534"

```

Seven check types are available:

Table 14: Assertion check types.

Check	Semantics
field_exists	Every matched node has the named field in its data_json
min_count	The matched set contains at least N items
max_count	The matched set contains at most N items
min_edges	Every matched node has at least N outgoing edges
unique_per_group	Within groups defined by group_by, the field value is unique
field_in_cidr	Every matched node's IP field is contained within the specified CIDR
custom_cel	Arbitrary CEL expression evaluated per matched entity; fails if false

This enables “policy as code” directly in the blueprint. Assertions are evaluated cross-layer (no layer filter), so a single assertion can validate properties across both physical and protocol layers.

4 Compilation Pipeline

NETCFG’s compilation pipeline has five distinct layers of abstraction, operating as a true network compiler. Figure 1 shows the data flow from abstract intent to concrete syntax.

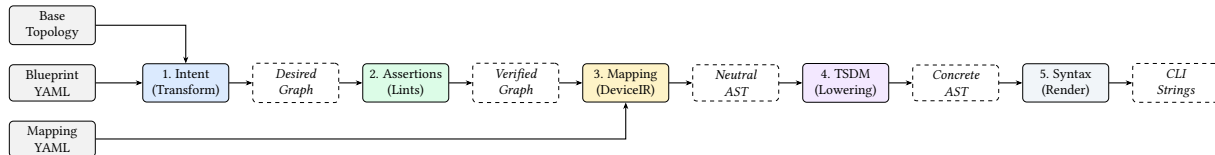


Figure 1: NetCfg compilation pipeline. High-level intent is progressively lowered through three distinct AST representations (Verified Graph, Neutral AST, Concrete AST) before serialization.

4.1 Layer 1: Intent Transformation

The blueprint is parsed into an AST and applied to a transactional shadow copy of the topology. Composable primitives (IPAM, Meshing, Overlays) mutate the graph to reach the desired state.

4.2 Layer 2: Topological Assertions

A DSL-driven assertion engine validates the desired graph against architectural invariants (e.g. area IDs must match, MTU consistency). Compilation aborts if invariants are violated.

4.3 Layer 3: DeviceIR Mapping

A Mapping DSL lowers the rich topology graph into a vendor-neutral **Abstract Syntax Tree (DeviceIR)**. This representation captures logical intent (e.g. "Neighbor X is remote-as Y") without hardware context.

4.4 Layer 4: Target-Specific Transform (TSDM)

The ****TSDM Layer**** performs AST-to-AST transformations. It lowers the Neutral AST into a platform-specific AST, remapping logical interfaces to physical ports (e.g. Eth1 → Eth1/1/1) based on the target’s chassis model and hardware layout.

4.5 Layer 5: Syntax Serialization

Jinja2 templates act as a dumb emission layer, serializing the final Target-Specific AST into vendor-specific CLI syntax. Zero logic is implemented at this layer.

The TransactionalOutput writer collects all files in temporary locations (using NamedTempFile in the same directory for same-filesystem guarantee), then atomically renames all via POSIX rename(2) on commit. If any stage fails, all temp files are dropped without committing—no partial output directory is ever visible.

Three artifacts are emitted per node:

- `{hostname}.conf`: the rendered device configuration.
- `{hostname}.deviceir.json`: the raw DeviceIR stanzas (audit).
- `{hostname}.lineage.json`: per-line mapping from config lines to the `rule_id` that produced them (provenance tracing).

4.5.1 Determinism guarantee

Given a fixed blueprint B , mapping M , and input topology G , the output configuration set $C = \text{compile}(G, B, M)$ is deterministic. Serialisation round-trip produces a bit-identical shadow. Primitives execute in declaration order. Selectors evaluate deterministically. Node ID allocation is monotonic. IP allocation iterates edges sorted by (src, dst) . DeviceIR stanzas use BTreeMap and are post-sorted. Template rendering is pure.

5 Core Mechanisms

5.1 Five-pass IPAM (`provision_ips`)

The IP allocator (`primitives/ipam.rs`) implements a five-pass algorithm:

1. **State discovery**: Scan all nodes for existing IP assignments (`src_ip`, `dst_ip`, `subnet` fields).
2. **Edge selection**: Evaluate the CEL selector; sort matched edges by (src, dst) for determinism.
3. **Pre-reservation**: Insert existing allocations into the allocator, preventing re-allocation and enforcing stickiness.
4. **Fulfilment**: For each edge, allocate a subnet using the configured strategy (dense, sparse, or hash). Extract host addresses from the subnet.
5. **Write-back**: Write `src_ip`, `dst_ip`, `subnet` (and IPv6 equivalents for dual-stack) into each endpoint's DeviceContext.

Identity-based stickiness. The allocator uses a canonical identity —`sorted(hostname_a, hostname_b)`—as the primary key. If a prior allocation exists for the same identity, the same prefix is returned without pool search. This anchors allocations to stable semantic identity, not integer IDs. If topology IDs change (e.g., after a re-import) but hostnames remain the same, allocations are preserved.

VRF isolation. The allocator maintains per-VRF state. Two primitives using the same CIDR pool in different VRFs can allocate overlapping subnets without conflict. Pool overlap *within* the same VRF is detected and rejected.

Allocation strategies:

- **dense**: Sequential allocation from the pool start. First edge gets the first available subnet.
- **sparse**: Skips every other subnet, leaving growth headroom between allocations.
- **hash**: Subnet position is derived from a hash of the edge identity, producing topology-independent allocations that do not depend on evaluation order.

Hierarchical pools. A `global_pool` primitive registers a large CIDR (e.g., `10.0.0.0/8`). Subsequent `provision_ips` primitives reference the pool by name and specify a `pool_size` (e.g., 24), carving /24 blocks from the parent pool on demand.

Dual-stack. When `ipv6_pool` is specified, both IPv4 and IPv6 addresses are allocated in a single primitive invocation. IPv6 fields (`src_ipv6`, `dst_ipv6`, `subnet_v6`) are written alongside IPv4 fields.

5.2 Protocol overlay construction (`build_protocol_layer`)

`build_protocol_layer` clones physical nodes into a named protocol layer. For each selected source-layer node:

1. Allocate a new monotonic node ID.
2. Add the node to the target layer.
3. Deep-merge the config overlay into the cloned node's metadata.
4. Record `_source_id` as a parent mapping to the physical node.
5. Optionally (`clone_underlying: true`) clone edges between source-layer nodes into the overlay.

Cross-layer traceability: the `_source_id` field links each protocol node to its physical origin. Since `provision_ips` already enriched the physical node's `DeviceContext` with IP addresses, the cloned protocol node inherits them via deep merge—this is how IP addresses flow from IPAM allocation to BGP configuration without explicit wiring.

Deep merge semantics: When both the base node and the config overlay contain nested JSON objects under the same key, the merge recurses key-by-key (overlay wins on conflict). This preserves sibling keys that the overlay does not mention. For example, a base node with `{bgp: {as: 65000, holdtime: 180}}` and an overlay with `{bgp: {timers: {keepalive: 60}}}` produces `{bgp: {as: 65000, holdtime: 180, timers: {keepalive: 60}}}`.

Idempotency: The primitive scans the target layer for existing nodes with `_source_id` matching source nodes. Already-cloned nodes are skipped, so re-running the primitive after a partial failure produces the correct result.

5.3 Configuration diffing (`DiffEngine`)

The diff engine (`diff/engine.rs`) computes a `MutationPlan` between two topologies by exporting each layer to JSON and comparing:

1. **Node diff:** Per layer, compare node sets by ID. Detect additions, deletions, and modifications (field-by-field property comparison via `extract_properties`, which unpacks the `data_json` blob).
2. **Edge diff:** Compare by `(src, dst)` pair with structural property comparison.

The `MutationPlan` contains six change types: `AddNode`, `RemoveNode`, `UpdateNode`, `AddEdge`, `RemoveEdge`, `UpdateEdge`. Changes are sorted deterministically by `(change_type, layer, id/src/dst)` for reproducible output.

The `plan` command displays the plan as a human-readable diff; the `ci-run` command uses it to compute per-device configuration diffs.

5.4 AST Transformations (TSDM Layer)

The **Target-Specific Device Model (TSDM)** layer implements the compiler's lowering phase via DSL-driven **AST-to-AST transformations**. This stage bridges the gap between logical design and physical hardware constraints.

5.4.1 The Transformation DSL

Lowering rules are defined in the blueprint using an expressive rewrite-rule grammar. Each rule consists of a `match_expr` (CEL predicate) and an `apply` map (field mutations).

Example: NX-OS interface lowering

```
transforms:
  - name: "nxos_lowering"
```

```

when: "device_os == 'nxos'"
rules:
  - match_expr: "kind == 'interface' && name.startsWith('Ethernet')"
    apply:
      name: "name + '/1'" # Ethernet1 -> Ethernet1/1
      mtu: "9216"         # Force jumbo frames

```

5.4.2 Execution Logic

The transformation engine (`dsl/transform.rs`) performs a deep-walk of the DeviceIR stanzas. For each stanza, it:

1. Binds the stanza's fields as CEL variables.
2. Evaluates the `match_expr` predicate.
3. If matched, executes the `apply` expressions to mutate the stanza's state.

This structured approach allows complex hardware logic—such as mapping logical ports to specific line-card slots in a modular chassis—to be expressed as programmable transformations rather than hardcoded logic or brittle templates.

The key architectural insight is that DeviceContext accumulates state across the entire primitive pipeline:

1. `mesh_nodes`: writes `mesh_type`, `peer_count`, and optionally `peer_interfaces` into metadata.
2. `provision_ips`: writes `src_ip`, `dst_ip`, `subnet`, `vrf`.
3. `build_protocol_layer`: deep-merges all of the above into the cloned protocol node, plus the config overlay.
4. `allocate_resources`: writes the allocated resource value (e.g., `asn`) into metadata.

By the time Stage 4 (rendering) occurs, each node's DeviceContext carries hostname, IPs, protocol metadata, interface assignments, and any custom fields—the complete information needed to produce a device configuration file.

6 CLI Reference

NETCFG provides eleven CLI commands spanning the development-to-deployment lifecycle. All commands use `clap` for argument parsing.

6.1 netcfg generate

Runs the full six-stage pipeline, writing artifacts to disk.

```

netcfg generate <topology> <blueprint> \
  --templates-dir <DIR> \
  [--output-dir <DIR>] [--emit-ir] [--verbose]

```

Argument	Default	Description
<code>topology</code>	—	Path to base topology archive (.nte)
<code>blueprint</code>	—	Path to blueprint YAML
<code>-templates-dir</code>	—	Directory containing .j2 templates
<code>-output-dir</code>	<code>./output</code>	Output directory for generated files
<code>-emit-ir</code>	<code>false</code>	Also write <code>{hostname}.ir.json</code>
<code>-verbose</code>	<code>false</code>	Print per-node rendering progress

Output: one .cfg file per device (plus .ir.json if `-emit-ir`). Absolute paths are printed to stdout, one per line, suitable for piping.

6.2 netcfg validate

Executes the pipeline in dry-run mode: all primitives run with `dry_run=true`, skipping DataFrame writes but verifying what *would* change. Optionally pre-flights templates by rendering to memory.

```
netcfg validate <topology> <blueprint> \
  [--templates-dir <DIR>] [--verbose]
```

Exit codes: 0 on success, non-zero on validation failure (parse errors, assertion failures, template syntax errors, IP pool conflicts).

6.3 netcfg plan

Computes the topology MutationPlan (current vs. desired) and displays additions, removals, and property changes. With `-diff`, it renders before/after configs and shows a unified text diff per device using the similar crate.

```
netcfg plan <topology> <blueprint> \
  [--diff] [--templates-dir <DIR>] [--verbose]
```

Output without `-diff`: colourised plan (green for additions, red for removals, yellow for modifications). Summary line: +N additions, -N removals across M layers.

Output with `-diff`: unified diff per device, preceded by `diff -cfg {hostname}`.

6.4 netcfg inspect

Provides three views into intermediate state:

```
netcfg inspect <path> --ast           # Print blueprint AST
netcfg inspect <path> --topology      # Export as Graphviz DOT
  [--blueprint <FILE>]
netcfg inspect <path> --trace <NODE_ID> # Trace node state changes
  --blueprint <FILE>
```

- `-ast`: Parses the file as a blueprint and prints the AST tree.
- `-topology`: Exports the graph as Graphviz DOT. With `-blueprint`, applies the blueprint first. Output includes per-layer subgraph clustering, hostname labels, and subnet annotations.
- `-trace <node_id>`: replays primitive execution and prints before/after DeviceContext snapshots for the specified node at each primitive step.

6.5 netcfg ci-run

The CI/CD integration point. Runs validate, plan, and config diff in a single invocation.

```
netcfg ci-run <topology> <blueprint> \
  --templates-dir <DIR> [--json]
```

With `-json`, emits a structured payload:

```
{
  "status": "success",
  "topology_additions": 42,
  "topology_removals": 0,
  "configuration_diffs": {
    "core-0": "- ip address 10.0.0.4/30\n+ ip address 10.0.0.8/30"
  },
  "warnings": []
}
```

This integrates with GitHub Actions, GitLab CI, or any CI system that can parse JSON.

6.6 netcfg import

Imports a YAML topology definition into an NTE archive.

```
netcfg import <yaml_path> <output.nt>
```

The importer reads a topogen-format YAML file (defining nodes with types, layers, and properties) and serialises it to an .nte archive that other commands consume.

6.7 netcfg lint

Runs design-rule assertions without the full validation pipeline. Applies the blueprint in dry-run mode, then evaluates assertions and reports results.

```
netcfg lint <topology> <blueprint> [--verbose]
```

Returns assertion results with pass/fail status, severity, and diagnostic messages. Exit code 1 only when error-severity assertions fail; warnings are reported but do not cause failure.

6.8 netcfg fmt

Standardises blueprint YAML formatting. Parses the blueprint via `Executor::parse()`, re-serialises with `serde_yaml`, and compares byte-for-byte. Idempotent.

```
netcfg fmt <blueprint> [--check] [--verbose]
```

With `-check`, exits with code 1 if formatting changes are needed (useful as a pre-commit hook or CI gate). Without `-check`, overwrites the file in place.

6.9 netcfg impact

Calculates the operational blast radius of a blueprint change against a base topology.

```
netcfg impact <topology> <blueprint> [--verbose]
```

For each mutation in the computed plan, assigns a risk level:

Risk	Colour	Trigger
High	Red	Node removal; critical property change (asn, router_id, src_ip, subnet); physical link add/remove
Medium	Yellow	Node addition; logical link add/remove
Low	Green	Metadata-only update

Critical properties are: `asn`, `router_id`, `src_ip`, `dst_ip`, `subnet`, `loopback`, `mgmt_ip`. Output includes per-node risk attribution with reasons.

6.10 netcfg report

Generates an automated Markdown architecture report from a compiled topology.

```
netcfg report <topology> <blueprint> -o <output.md> [--verbose]
```

The report contains three sections:

1. **Node inventory:** hostname, type, OS, ASN, loopback, management IP.
2. **Physical links & IP addressing:** source/destination hosts, interfaces, IPs, subnets.

3. **Topology visualisation:** A `netviz-json` code block with a custom JSON schema (`NetVizNode`, `NetVizEdge`, `NetVizTopology`) suitable for rendering by external visualisation tools.

6.11 netcfg export-clab

Generates a Containerlab `clab.yaml` simulation file from a compiled topology, enabling rapid lab deployment for testing (e.g. the three-site mesh from §7).

```
netcfg export-clab <topology> <blueprint> \
  -o <clab.yaml> [--configs_dir <DIR>] [--verbose]
```

Heuristically maps `device_os` to Containerlab container kinds: `cisco_iosxr` → `vr-xrv9k`, `nokia_srl` → `srl`, `arista_eos` → `ceos`, `default` → `linux`. When `--configs_dir` is provided, bind-mounts generated configuration files into OS-specific remote paths (e.g. `/etc/opt/srlinux/config.json` for SRL, `/mnt/flash/startup-config` for cEOS).

7 End-to-End Worked Example

We trace the `three-site-mesh.yaml` blueprint through every pipeline stage. The base topology has 12 routers across 3 sites (4 per site: `site-a-r1` through `site-c-r4`).

7.1 Input: Blueprint

Listing 5: `three-site-mesh.yaml` (abridged).

```
version: 1
layers:
  # Stage 1: Full mesh within each site
  - name: input
    primitives:
      - type: mesh_nodes
        selector: "nodes[site='a']"
        mesh_type: full
      - type: mesh_nodes
        selector: "nodes[site='b']"
        mesh_type: full
      - type: mesh_nodes
        selector: "nodes[site='c']"
        mesh_type: full

  # Stage 2: IP allocation per site
  - name: input
    primitives:
      - type: provision_ips
        selector: "edges[src>=0]"
        pool: "10.1.0.0/24"
        subnet_size: 30
        strategy: dense

  # Stage 3: BGP overlay
  - name: input
    primitives:
      - type: build_protocol_layer
        selector: "nodes[true]"
        layer: bgp
        config:
          protocol_type: bgp
          asn_base: "65000"
```

7.2 Stage 1: Parse and validate

The parser validates the YAML, confirms `version: 1`, and compiles each selector string (`nodes[site='a']`, `edges[src>=0]`, etc.) to a CEL program. The normaliser converts `site='a'` to `site=="a"`.

7.3 Stage 2: Shadow topology

The executor clones the base topology via a serialisation round-trip, creating an isolated shadow copy. All subsequent mutations operate on the shadow; the base topology is never modified.

7.4 Stage 3: Primitive execution

mesh_nodes (three invocations): Creates $3 \times \binom{4}{2} = 18$ edges (6 per site). The selector nodes[site='a'] matches the 4 nodes with {site: "a"} in their data_json. Each node gets mesh_type: "full" and peer_count: 3 written to its metadata.

provision_ips: The selector edges[src>=0] matches all 18 edges. Edges are sorted by (src, dst). The five-pass algorithm allocates /30 subnets from 10.1.0.0/24:

- Edge (1,2): subnet 10.1.0.0/30, IPs 10.1.0.1 and 10.1.0.2
- Edge (1,3): subnet 10.1.0.4/30, IPs 10.1.0.5 and 10.1.0.6
- ...continuing through all 18 edges.

After this stage, node site-a-r1 has: src_ip: "10.1.0.1", subnet: "10.1.0.0/30", vrf: "default".

build_protocol_layer: Clones all 12 physical nodes into the bgp layer (new IDs 13–24). Each clone receives:

- protocol_type: "bgp" and asn_base: "65000" (from the config overlay)
- _source_id: 1 (parent mapping)
- All properties from the physical node, including the IPs written by provision_ips (via deep merge)

7.5 Stage 4: Mapping evaluation

The companion mapping (three-site-mesh-mapping.yaml):

```
rules:
- selector: "all_nodes"
  stanza:
    kind: "bgp"
    fields:
      node: "site-a-r1"
      local_asn: "65001"
      description: "site-a full-mesh BGP session"
```

The evaluator matches all nodes and emits one stanza per match. Each stanza carries provenance: {rule_id: "rule_0"}.

7.6 Stage 5: Render

Stanzas are grouped by fields.node (all mapped to site-a-r1 in this example). The fallback template produces:

```
kind=bgp key=
```

7.7 Stage 6: Atomic write

Three files are written atomically:

- site-a-r1.conf: the rendered configuration
- site-a-r1.deviceir.json: the raw stanza list
- site-a-r1.lineage.json: per-line provenance

8 Error Model

NETCFG uses typed error enums with `thiserror` derivation and `miette` diagnostic annotations. Table 15 summarises the error types.

Table 15: Error enums and their variants.

Enum	Variants	Context
<code>ParseError</code>	<code>YamlSyntax</code> , <code>Validation</code> , <code>Deserialize</code> , <code>LayerOrdering</code>	Blueprint parsing
<code>SelectorError</code>	<code>InvalidSyntax</code> , <code>Compile</code> , <code>Evaluate</code> , <code>NonBooleanResult</code> , <code>Polars</code> , <code>Json</code> , <code>TopologyAccess</code>	Selector compilation and evaluation
<code>PrimitiveError</code>	<code>Validation</code> , <code>Execute</code> , <code>Selector</code> , <code>Topology</code> , <code>Graph</code> , <code>Polars</code> , <code>Json</code>	Primitive execution
<code>ExecuteError</code>	<code>Parse</code> , <code>Primitive</code> , <code>Shadow</code> , <code>Plan</code> , <code>Assertion</code>	Top-level executor
<code>ShadowError</code>	<code>Clone</code> , <code>Commit</code> , <code>Validation</code>	Shadow topology lifecycle
<code>AssertionError</code>	<code>Failed</code> , <code>Selector</code>	Assertion evaluation
<code>DiffError</code>	<code>TopologyAccess</code> , <code>ParseError</code>	Configuration diffing
<code>RenderError</code>	<code>Io</code> , <code>Jinja</code> , <code>Json</code> , <code>MissingTemplate</code>	Template rendering

`ExecuteError` variants carry `miette::Diagnostic` annotations with diagnostic codes (e.g., `netcfg::parse`, `netcfg::primitive::execution_failed`, `netcfg::assertion`) enabling structured error handling in CI pipelines.

`ParseError::YamlSyntax` includes a source span (`miette::SourceSpan`) pointing at the offending line, enabling IDE-quality error rendering.

9 Editor and Service Integration

9.1 Language Server (LSP)

The `netcfg-lsp` crate provides a Language Server Protocol implementation built on `tower-lsp` and `Tokio`. It communicates via `stdin/stdout` and offers three capabilities:

- Real-time diagnostics.** On file open, change, and save, the server calls `parse_blueprint()` and maps errors to LSP diagnostics with line-column precision. For `serde_yaml` errors (which embed location as free text), a regex extracts “at line N column M”. Diagnostics are published with source label `ank_netcfg` and severity `ERROR`.
- Primitive auto-completion.** When the cursor follows a `type:` token (detected by a suffix heuristic), the server offers completion items for the six core primitives, each with a one-line docstring. Trigger characters are space and colon.
- Document synchronisation.** Open documents are tracked in a `DashMap<String, String>` (lock-free concurrent map). Full-text sync (`TextDocumentSyncKind::FULL`) is used for correctness. On close, the document is removed and diagnostics are cleared.

9.2 REST API Microservice

The `netcfg-api` crate exposes the compiler as a stateless HTTP service via `axum`, enabling CI/CD pipelines and enterprise orchestrators to compile blueprints without a local `netcfg` binary.

Endpoint: `POST /compile (port 3000)`.

Table 16: Compile endpoint request and response schema.

Direction	Field	Type	Description
Request	topology_zip_base64	String	Base64-encoded .nte archive
Request	blueprint_yaml	String	Raw YAML blueprint
Response	status	String	success or error
Response	configs	Map<String, String>	Hostname → rendered config
Response	device_ir	Map<String, JSON>	Hostname → DeviceContext IR
Response	warnings	Vec<String>	Compilation warnings

The compilation pipeline decodes the base64 payload into a temporary file, loads the topology, parses the blueprint, executes via `Executor::apply()`, and extracts per-device IR. All processing is in-memory with no persistent state; the service scales horizontally behind a load balancer.

10 Limitations

10.1 edge_properties not applied (mesh_nodes)

Description. The `MeshNodesSpec.edge_properties` field is accepted by the blueprint parser but intentionally not applied. Applying it requires an edge `DataFrame` API on `nte_topology::Topology` equivalent to `set_dataframe/update_dataframe` but keyed on `(source_id, target_id)` pairs.

Workaround. Edge properties can be encoded as node metadata on both endpoints (e.g., `link_bandwidth` as a per-node field keyed by peer ID).

Planned resolution. Awaiting `nte-topology` ≥ 0.2 edge `DataFrame` support. A tracking test (`edge_properties_deferred_until_nte_edge_dataframe`) is present in the test suite.

10.2 Mapping AST inspection not implemented

Description. The `inspect -ast` command can parse and display blueprint ASTs but does not yet support mapping documents. Attempting to inspect a mapping file prints “Mapping AST visualization coming soon.”

Workaround. Inspect the mapping YAML directly. The format is simple (a rules list with `selector` and `stanza` fields).

Planned resolution. When the mapping DSL grows beyond simple selector/stanza pairs, AST visualization will be added.

10.3 Shadow validation is a placeholder

Description. The `ShadowTopology::validate()` method currently only checks that the shadow did not become empty. It does not perform structural integrity checks (dangling edges, orphaned protocol nodes, etc.).

Workaround. Design-rule assertions can catch many structural problems. Use `min_edges` and `field_exists` assertions as guardrails.

Planned resolution. Integrate with the NTE policy validation framework when available.

10.4 No intra-primitive rollback

Description. If a primitive fails partway through execution (e.g., `provision_ips` exhausts its pool after allocating half the edges), the shadow topology retains the partial mutations. The pipeline aborts (returning an error), and the shadow is discarded, but there is no mechanism to roll back individual primitives within a shadow.

Workaround. This is usually not a problem: the entire shadow is discarded on error, so no partial state reaches production. The limitation matters only for debugging: the error message may not reflect the full extent of partial mutations.

Planned resolution. No current plan. Intra-primitive rollback would require snapshotting the shadow before each primitive, which doubles memory usage.

10.5 generate CLI bypasses mapping document

Description. The generate CLI command renders configs directly from DeviceContext via templates, bypassing the mapping document. The full six-stage pipeline (with mapping evaluation and DeviceIR) is implemented in `config_gen::generate_configs()` but the CLI does not expose it yet.

Workaround. Use the `generate_configs()` function directly from Rust code for the full pipeline with mapping support.

Planned resolution. Add `-mapping` flag to the generate CLI command.

10.6 No incremental compilation

Description. Every invocation recompiles the entire topology from scratch. The diff engine can identify which devices changed, but the pipeline cannot skip unchanged devices during compilation.

Workaround. Use `ci-run -json` to compute the diff, then selectively push only the changed configuration files.

Planned resolution. Under investigation. Incremental compilation requires caching intermediate state (the desired topology) between runs and detecting which blueprint changes invalidate which nodes.

11 Future Work

- **GPU-accelerated primitive execution:** Offloading selector evaluation and IPAM allocation to the GPU via NTE's WebGPU compute backend, targeting 100K+ node topologies.
- **WASM linear memory I/O:** The current `wasm_plugin` primitive validates module compilation and symbol resolution but mocks the JSON exchange via linear memory. Full C-ABI memory allocation between the Rust host and WASM guest is required for production use.
- **API template rendering:** The `POST /compile` endpoint currently returns placeholder configuration text; integrating the MiniJinja rendering engine into the API service is planned.
- **Mapping CLI integration:** Exposing the full six-stage pipeline (including mapping evaluation) through the generate CLI command.
- **Incremental compilation:** Caching intermediate state to avoid recompiling unchanged portions of the topology.
- **LSP assertion integration:** Extending the language server to evaluate design-rule assertions in real time, providing inline warnings for selector violations and IP pool capacity.

12 References

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A Blueprint YAML Schema Reference

Listing 6: Complete blueprint schema (annotated).

```

version: 1                                # integer, >= 1
imports: [<path>, ...]                  # optional, relative file paths to merge

layers:
- name: <string>                          # non-empty, layer identifier
  requires: [<string>, ...]              # optional, previously-declared layers
  primitives:

    # --- mesh_nodes ---
    - type: mesh_nodes
      selector: <cel_selector>
      mesh_type: full | hub_and_spoke | route_reflector
      hub_selector: <cel_selector>        # required for h&s
      spoke_selector: <cel_selector>     # required for h&s
      edge_properties: <json_object>    # deferred
      naming_strategy: eth | cisco_ge | junos_ge

    # --- provision_ips ---
    - type: provision_ips
      selector: <cel_selector>
      pool: <cidr_or_pool_name>
      subnet_size: <u8>                  # default: 30 (v4), 126 (v6)
      ipv6_pool: <cidr_or_pool_name>    # optional dual-stack
      ipv6_subnet_size: <u8>
      strategy: dense | sparse | hash   # default: dense
      pool_size: <u32>                 # for global pool carving
      vrf: <string>                    # default: "default"

    # --- build_protocol_layer ---
    - type: build_protocol_layer
      selector: <cel_selector>
      layer: <string>                  # target layer
      config: <json_object>           # deep-merged into clone
      clone_underlying: <bool>        # default: false

    # --- allocate_resources ---
    - type: allocate_resources
      selector: <cel_selector>
      resource_type: <string>
      pool: <range_or_pool_name>
      strategy: dense                  # default: dense
      pool_size: <u32>

    # --- global_pool ---
    - type: global_pool
      name: <string>
      pool: <cidr>
      vrf: <string>

    # --- global_resource_pool ---
    - type: global_resource_pool
      name: <string>
      resource_type: <string>
      pool: <range>

    # --- conditional ---
    - type: conditional
      condition: <cel_expression>
      then_primitives: [<primitive>, ...]
      else_primitives: [<primitive>, ...] # optional

    # --- custom_primitive ---

```

```

- type: custom_primitive
  name: <string>
  parameters: {<string>: <json_value>}
  primitives: [<primitive>, ...]

# --- inject_secrets ---
- type: inject_secrets
  selector: <cel_selector>
  secrets: {<metadata_key>: <env_var_name>}

# --- build_routing_policy ---
- type: build_routing_policy
  selector: <cel_selector>
  name: <string> # policy name
  statements:
    - name: <string> # sequence number
      action: permit | deny
      match_condition: <string> # optional
      set_action: <string> # optional

# --- build_access_policy ---
- type: build_access_policy
  selector: <cel_selector>
  name: <string> # policy name
  rules:
    - name: <string>
      action: permit | deny
      source: <string> # optional
      destination: <string> # optional
      protocol: <string> # optional
      port: <string> # optional

# --- wasm_plugin (requires --features wasm) ---
- type: wasm_plugin
  selector: <cel_selector>
  plugin_path: <filesystem_path> # path to .wasm binary

assertions:
- name: <string>
  severity: error | warning
  select: <cel_selector>
  check:
    field_exists: <field_name>
    # OR
    min_count: <integer>
    # OR
    max_count: <integer>
    # OR
    min_edges: <integer>
    # OR
    unique_per_group:
      field: <field_name>
      group_by: <field_name>
    # OR
    field_in_cidr:
      field: <field_name>
      cidr: <cidr>
    # OR
    custom_cel:
      expression: <cel_expression>

```

B Selector DSL Grammar

Listing 7: Selector DSL grammar (informal EBNF).

```

selector    ::= "all()" | target "[" filter "]"
target      ::= "nodes" | "edges"
filter      ::= cel_expression

(* Surface syntax normalisations applied before CEL compilation *)
"="         -> "=="
"and"       -> "&&"
"or"        -> "||"
"not"       -> "!"
"&"        -> "&&"
"|"        -> "||"

(* CEL variables bound per entity *)
nodes: id (int), type (string), layer (string), + all data_json fields
edges: id (string), src (int), dst (int), layer (string),
      + all data_json fields

(* IP prefix length extraction *)
For any field with an IP/CIDR value "10.0.0.0/30":
  {field}_len = 30 (integer, bound automatically)

(* Undeclared references *)
Accessing a field that does not exist on a node evaluates to false
(not an error), enabling heterogeneous topologies.

```

C CLI Quick Reference

Table 17: CLI command summary.

Command	Synopsis
generate	Full pipeline: parse → transform → render → write
validate	Dry-run pipeline with optional template pre-flight
plan	Compute and display topology mutation plan (or config diff)
inspect	AST visualisation, Graphviz DOT export, or per-node trace
ci-run	Combined validate + plan + config diff for CI/CD
import	Import YAML topology to NTE archive
lint	Run design-rule assertions only
fmt	Standardise blueprint YAML formatting
impact	Calculate blast radius of a blueprint change
report	Generate Markdown architecture report
export-clab	Generate Containerlab simulation file